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A High-Order Radial Basis Function Finite Difference Method for Linear Anisotropic Elliptic Interface Problems

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Abstract
Anisotropic elliptic interface problems are prevalent in various scientific and engineering disciplines. Traditional mesh-based numerical methods often struggle with complex geometries near interfaces, which can significantly impact their performance. In contrast, meshless methods, which eliminate the need for mesh generation, offer inherent advantages in handling interface problems and have emerged as a promising alternative. In this paper, we propose a high-order radial basis function finite difference (RBF-FD) method, which is specially designed to solve second-order linear anisotropic elliptic interface problems. The main idea of our method is to utilize the domain decomposition associated with the interfaces and construct the RBF-FD approximations of functions and differential operators in each of the subdomains independently to resolve inherent discontinuities on the interfaces. Detailed discretization schemes for the proposed method are presented together with implementation algorithm. Extensive numerical experiments on various interfaces are also performed to demonstrate high-order accuracy and computational efficiency of the proposed RBF-FD method.

Keywords Linear elliptic interface problem · Anisotropic tensor · Radial basis function · Finite difference · Subdomain-based stencil

Mathematics Subject Classification 65N22 · 80M22 · 35K57

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1 Introduction

An steady-state interface problem on a bounded domain $\Omega \subset \mathbb{R}^d$ with boundary $\partial\Omega$ is often described in the following form:

$$\mathcal{L}(u) = f, \quad \boldsymbol{x} \in \cup_{i=1}^{K} \Omega_i, \tag{1a}$$

$$[\![u]\!] = h_{ij}, \quad \boldsymbol{x} \in \{\Gamma_{ij}\}, \tag{1b}$$

$$[\![\mathcal{J}(u) \cdot \boldsymbol{n}_{ij}]\!] = q_{ij}, \quad \boldsymbol{x} \in \{\Gamma_{ij}\}, \tag{1c}$$

$$u = g, \quad \boldsymbol{x} \in \partial\Omega, \tag{1d}$$

where the domain Ω is partitioned into K open subdomains $\{\Omega_1, \Omega_2, \ldots, \Omega_K\}$ by the closed interfaces $\{\Gamma_{i,j} = \Omega_i \cap \Omega_j \neq \emptyset \,|\, i \neq j\}$, the notation $[\![v]\!]$ represents the jump of v across the interface $\Gamma_{i,j}$, i.e.,

$$[\![v]\!] = v|_{\Omega_i} - v|_{\Omega_j},$$

the vector $\boldsymbol{n}_{i,j}$ denotes the unit outward normal vector on $\Gamma_{i,j}$ from Ω_i to Ω_j, and the functions f, g, h_{ij} and q_{ij} represent the source term, the boundary value, and the two types of jump vlaues, respectively. The operators $\mathcal{L}$ and $\mathcal{J}$ usually act on differential forms within the subdomains and across the interfaces, respectively. In this paper, we focus on the second-order linear anisotropic elliptic interface problem, in which $\mathcal{L}$ denotes the classic linear anisotropic diffusion reaction operator, i.e.,

$$\mathcal{L}(u) = -\nabla \cdot (\Lambda \nabla u) + \sigma u, \tag{2}$$

and correspondingly

$$\mathcal{J}(u) = \Lambda \nabla u. \tag{3}$$

Here Λ is the symmetric positive definite diffusion tensor, meaning that there exists a positive constant c_0 such that

$$\boldsymbol{\xi}^\top \Lambda \boldsymbol{\xi} \geq c_0 > 0, \quad \forall\, \boldsymbol{\xi} \neq \mathbf{0},$$

and $\sigma > 0$ is the reaction coefficient, which is required to be nonnegative. Furthermore, both σ and Λ exhibit discontinuities across the interface Γ, i.e.,

$$\sigma(\boldsymbol{x}) = \sigma_k(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega_k \tag{4}$$

and

$$\Lambda(\boldsymbol{x}) = \Lambda_k(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega_k \tag{5}$$

for $k = 1, \cdots, K$. It should be noted that the numerical method developed in this paper are also applicable to more general elliptic interface problems.

Elliptic interface problems with discontinuous coefficients play a crucial role in various science and engineering applications, such as composite materials and fluid dynamics. Significant progress has been made in the development of numerical methods for interface problems in the past decades. The immersed interface method (IIM) proposed by LeVeque et al. in [18] has become a popular approach for solving linear elliptic PDE problems with discontinuous scalar-values diffusion coefficients. This method has been seamlessly integrated into spatial discretization techniques such as finite difference methods (FDM), finite element methods (FEM), and finite volume methods (FVM). Subsequently, substantial developments along this direction have been made, with notable examples including the second-order explicit-jump IIM [35], the third-order 9-point compact FDM [23], and the fourth-order IIM [41] for elliptic interface problems. At the same time, the elliptic interface problems with discontinuous anisotropic diffusion tensors pose more numerical challenges, and many excellent

works also have been emerged to address such problems. For example, Pan et al. [24] developed an easily implementable exact-interface-fit mesh generation algorithm and along with it proposed a linearity-preserving finite volume method. Dong et al. [6] introduced a novel finite element-finite difference (FE-FD) method that combines a finite element discretization (away from the interface) with a finite difference discretization (near or on the interface) for solving these problems. The traditional numerical methods for solving interface problems usually require meshes that are precisely fitted to the interfaces to achieve acceptable approximation accuracy. However, generating such interface-fitting meshes presents a significant challenge, especially in three-dimensional problems or those involving evolving interfaces. To address this challenge, some unfitted methods also have been developed, such as the generalized/expanded finite element method [17] and the cut finite element method [13]. These methods have been carefully designed to alleviate the intricacy of mesh generation, offering more efficient and practical alternatives for solving interface problems.

In recent years, significant advancements have been made in the application of meshless methods to interface problems. As outlined in [30], meshless methods for solving partial differential equations (PDEs) can be broadly classified into two categories: one based on the fitting principle, and the other based on the interpolation principle. Deep learning methods, such as the physics-informed neural network (PINN, [25]) and the deep Ritz method (DRM, [8]), fall into the former category. These methods have been effectively applied to solve interface problems. For instance, [7] introduced static weights into the loss function and designed an adaptive sampling strategy to enhance the accuracy of deep learning methods for solving interface problems. A similar approach is seen in [37]. Fitting-based methods are ultimately reduced to solving a least squares problem, with notable examples including TransNet [40], the random feature method (RFM) [4]), and the extreme learning machine (ELM) [14]). Liang et al. [19] employed the ELM method to address interface problems and achieved exceptional accuracy. Additionally, the high-order generalized finite difference Method (GFDM) recently proposed by [38] also belongs to this category. This work, based on Taylor series expansion and weighted moving least square technique, has been successfully applied to solving anisotropic elliptic problems.

Radial basis function (RBF) methods are a prominent class of meshless techniques for solving PDEs. Key examples include the RBF collocation method [16], the RBF Galerkin method [32], and the RBF finite difference (RBF-FD) method [9]. The RBF collocation method, a fitting-based approach first introduced by [16], encounters challenges as the dimensionality increases, leading to dense linear systems and significant computational costs. Localized RBF approximation methods, such as the RBF-FD method, have been developed to effectively address these limitations. The RBF-FD method, first proposed by [30], is an interpolation-based approach that produces sparse linear systems. Natasha et al. further enhanced the RBF-FD method [9], improving its accuracy and stability. The RBF-FD method has since found extensive applications in solving a variety of practical problems. These include the overlapping RBF-FD method [26], the RBF-FD for Navier-Stokes equations [5], the RBF-FD for 3D magnetohydrostatics [22], the RBF-FD for surface equations [34], and numerous others [2, 9, 29].

For interface problems, discontinuous coefficients and jump conditions substantially increase the complexity of discretization and numerical solution. To address these challenges, a variety of improved radial basis function (RBF)-based methods have been proposed in recent years. Within the global collocation framework, existing studies primarily focus on domain decomposition, explicit enforcement of jump conditions, and careful adjustment of shape parameters. In one dimension, Shin and Jung [27] proposed a spectral RBF collocation method for elliptic interface problems with discontinuous coefficients or singular source

terms. The computational domain is partitioned into multiple subdomains, within which local collocation solutions are constructed independently; weighted norms are employed in the spectral component, while a weighted shape-parameter strategy is adopted for the RBF component to enhance numerical stability. For two-dimensional interface problems, Gholampour et al. developed an RBF-PUM method equipped with RBF-QR stabilization [10, 11]. This approach combines domain decomposition with local RBF-QR approximations on each subdomain and explicitly enforces jump conditions along interfaces. The RBF-QR technique expands Gaussian radial basis functions into truncated series and applies QR factorization to the resulting interpolation matrix, thereby effectively alleviating the severe ill-conditioning that arises when the shape parameter ε is small. In addition, Yazdani et al. [39] introduced an adaptive shape-parameter strategy that dynamically adjusts shape parameters in the vicinity of interfaces according to local solution features. Several other RBF collocation methods have also been proposed for problems involving discontinuities [1, 28]. Overall, these methods mitigate numerical ill-conditioning and suppress nonphysical oscillations near interfaces to a certain extent through localization and shape-parameter control. Nevertheless, they generally rely on solving dense, overdetermined linear systems via least-squares, only partially reduce the sensitivity to shape parameters, and may still exhibit Gibbs-type oscillations, particularly in the presence of high coefficient contrasts or strong discontinuities.

At the same time, localized RBF methods have also been extended to interface problems. For instance, the radial basis function-generated finite difference (RBF-FD) method has been applied to heat equilibrium problems in domains with heat-transfer interfaces [20]. However, such methods are typically restricted to homogeneous jump conditions and require explicit enforcement of interface conditions, which makes the treatment of sharp interfaces challenging. Moreover, their performance strongly depends on the choice of Gaussian basis functions, and the associated shape parameters often require careful, problem-dependent tuning. In this paper, we propose a high-order RBF-FD method for second-order linear anisotropic elliptic interface problems, employing polyharmonic splines as the radial basis functions. The method is completely shape-parameter-free, generates sparse linear systems, and achieves high-accuracy approximations of both the solution and flux through unified jump-condition enforcement, providing a robust, efficient, and easily extendable alternative for interface problems.

The remainder of the paper is organized as follows: Section 2 briefly reviews the RBF interpolation and the RBF-FD method. Section 3 presents the proposed high-order RBF-FD method with detailed discretization schemes and implementation algorithm. Numerical experiments in two and three dimensions are provided in Section 4 to demonstrate high-order accuracy and efficiency of the proposed method. Finally, some conclusions are summarized in Section 5.

2 Preliminaries on Radial Basis Function Methods

The radial basis function (RBF) is a widely used tool in function approximation, interpolation, and machine learning. It is a scalar function depending on the distance to a center point $\boldsymbol{c} \in \mathbb{R}^d$ and is typically defined as:

$$\Phi(\boldsymbol{x}) = \phi(\|\boldsymbol{x} - \boldsymbol{c}\|), \quad \forall\, \boldsymbol{x} \in \mathbb{R}^d$$

Table 1 A list of commonly used RBFs, where $r = \|\boldsymbol{x} - \boldsymbol{c}\|$, $\epsilon > 0$ is the shape parameter, p is the certain polynomials, and K_m denotes the Bessel K-function.

Type of basis function	RBF $\phi(r)$
Gaussian	$\exp(-\varepsilon r^2)$
MQ (Multiquadric)	$\sqrt{1+\varepsilon r^2}$
IMQ (Inverse MQ)	$1/\sqrt{1+\varepsilon r^2}$
IQ (Inverse quadratic)	$1/1+\varepsilon r^2$
Wendland [31]	$(1-\varepsilon r)_+^m p(\varepsilon r)$
Matern [21]	$\frac{2^{1-m}}{\Gamma(m)} r^m K_m(\varepsilon r), m > 0$
Polyharmonic Spline (PHS)	r^{2m+1} or $r^{2m}\log r$, $m \in \mathbb{N}$

where $\phi : [0, \infty) \to \mathbb{R}$, $\boldsymbol{x}$ represents the input point and $\|\cdot\|$ denotes the Euclidean distance. There are many choices for radial basis functions and Table 1 provides a list of commonly used ones.

2.1 RBF interpolation

Given a set of scattered data $\{(\boldsymbol{x}_i, f_i)\}_{i=1}^n$ with $\boldsymbol{x}_i \in \mathbb{R}^d$ and $f_i \in \mathbb{R}$, the goal of RBF interpolation is to find a function $s(\boldsymbol{x})$ in the form of

$$s(\boldsymbol{x}) = \sum_{i=1}^{n} \lambda_i \phi(\|\boldsymbol{x} - \boldsymbol{x}_i\|), \tag{6}$$

such that

$$s(\boldsymbol{x}_j) = f_j, \quad j = 1, 2, \ldots, n. \tag{7}$$

Clearly, solving the RBF interpolation problem in (7) is equivalent to solving the following linear system:

$$\begin{pmatrix} \phi_{11} & \cdots & \phi_{1n} \\ \vdots & & \vdots \\ \phi_{n1} & \cdots & \phi_{nn} \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix}, \tag{8}$$

where

$$\phi_{ij} = \phi(r_{ij}) \tag{9}$$

with $r_{ij} = \|\boldsymbol{x}_j - \boldsymbol{x}_i\|$.

It is noteworthy that for certain choices of RBF (such as the Gaussian function), the coefficient matrix in (8) can be guaranteed to be non-singular. However, for the majority of RBFs, the coefficient matrix is usually non-invertible [3, 15, 33], which greatly limits the applicability of RBF interpolation. To overcome this limitation, the concept of conditionally positive definite (CPD) RBFs [33] was further developed.

Definition 1 A RBF ϕ is called conditionally positive definite of order p in $\mathbb{R}^d$ if

$$\sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i \lambda_j \phi(\|\boldsymbol{x}_i - \boldsymbol{x}_j\|) > 0, \tag{10}$$

for any n pairwise distinct points $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n \in \mathbb{R}^d$ and $\boldsymbol{\lambda} = (\lambda_1, \cdots, \lambda_n)$ satisfying

$$\sum_{i=1}^{n} \lambda_i \psi_j(\boldsymbol{x}_i) = 0, \quad j = 1, 2, \cdots, m, \tag{11}$$

for any basis functions $\{\psi_j\}_{j=1}^m$ of the space of polynomials of degree at most p.

Note that m is the dimension of the space of polynomials of degree at most p in (11), and we have

$$m = C_{p+d}^d = \frac{(p+d)!}{p!d!}.$$

For example, let us consider the space of polynomials in $\mathbb{R}^2$ ($d = 2$). If we select $p = 2$, then the dimension of the corresponding polynomial space is $m = 6$, and a set of basis functions can be given by $\psi_1 = 1$, $\psi_2 = x$, $\psi_3 = y$, $\psi_4 = x^2$, $\psi_5 = xy$ and $\psi_6 = y^2$. Since the polyharmonic spline (PHS) has been proven to be conditionally positive definite, we will only consider using them as our RBF ϕ in this paper. We now define the modified interpolation function

$$s(\boldsymbol{x}) = \sum_{i=1}^{n} \lambda_i \phi(\|\boldsymbol{x} - \boldsymbol{x}_i\|) + \sum_{j=1}^{m} \alpha_j \psi_j(\boldsymbol{x}), \tag{12}$$

and by additionally imposing the constraints (11), the CPD-RBF interpolation problem can be formulated as follows:

$$\begin{cases} \displaystyle\sum_{j=1}^{n} \lambda_j \phi_{ij} + \sum_{j=1}^{m} \alpha_j \psi_{ij} = f_i, \quad i = 1, 2, \cdots, n, \\ \displaystyle\sum_{i=1}^{n} \lambda_i \psi_{ij} = 0, \quad j = 1, 2, \cdots, m, \end{cases} \tag{13}$$

where ϕ_{ij} is defined in (9) and

$$\psi_{ij} = \psi_j(\boldsymbol{x}_i). \tag{14}$$

Its matrix form is then given by

$$\begin{pmatrix} \phi_{11} & \cdots & \phi_{1n} & \psi_{11} & \cdots & \psi_{1m} \\ \vdots & & \vdots & \vdots & & \vdots \\ \phi_{n1} & \cdots & \phi_{nn} & \psi_{n1} & \cdots & \psi_{nm} \\ \psi_{11} & \cdots & \psi_{1n} & 0 & & 0 \\ \vdots & & \vdots & \vdots & & \vdots \\ \psi_{1m} & \cdots & \psi_{nm} & 0 & & 0 \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \\ \alpha_1 \\ \vdots \\ \alpha_m \end{pmatrix} = \begin{pmatrix} f_1 \\ \vdots \\ f_n \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{15}$$

To guarantee the invertibility of (15), it is necessary to ensure that the upper right block $(\psi_{ij})_{n\times m}$ has full column rank, i.e., $n \geq m$, as shown in [36].

2.2 RBF Finite Difference Approximation for Differential Operators

In this section, we explore the construction of the RBF finite difference (RBF-FD) scheme for differential operators. Given a set of scattered points, we will discuss, without loss of generality, how to construct the RBF-FD approximation of a differential operator $\mathcal{D}(u)$ at a specific point $\boldsymbol{x}_1$. Let the n nearest neighbors of $\boldsymbol{x}_1$ be $\{\boldsymbol{x}_i\}_{i=1}^n$ (including $\boldsymbol{x}_1$ itself, also

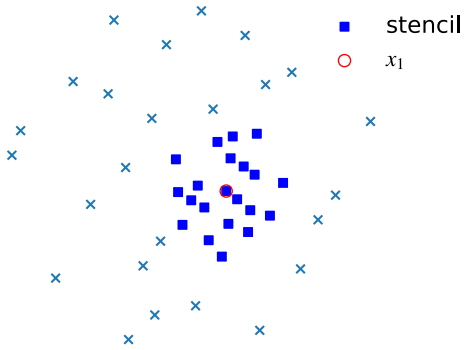


Fig. 1 The stencil for the point $\boldsymbol{x}_1$.

known as stencil, see Figure 1 for an illustration in two dimensions). The value of $\mathcal{D}(u)$ at $\boldsymbol{x}_1$ can be approximated as a linear combination of the function values at these neighboring points, i.e.,

$$\mathcal{D}(u)(\boldsymbol{x}_1) \approx \sum_{k=1}^{n} w_k u(\boldsymbol{x}_k). \tag{16}$$

The next step is to determine the combination coefficients $\{w_k\}_{k=1}^n$ in (16). By using the interpolation (12) to replace u in (16) , we have

$$\sum_{i=1}^{n} \lambda_i \mathcal{D}(\phi)(\|\boldsymbol{x}_1 - \boldsymbol{x}_i\|) + \sum_{j=1}^{n} \alpha_j \mathcal{D}(\psi_j)(\boldsymbol{x}_1) = \sum_{i=1}^{n} \lambda_i \sum_{k=1}^{n} w_k \phi_{ik} + \sum_{j=1}^{n} \alpha_j \sum_{k=1}^{n} w_k \psi_{kj}. \tag{17}$$

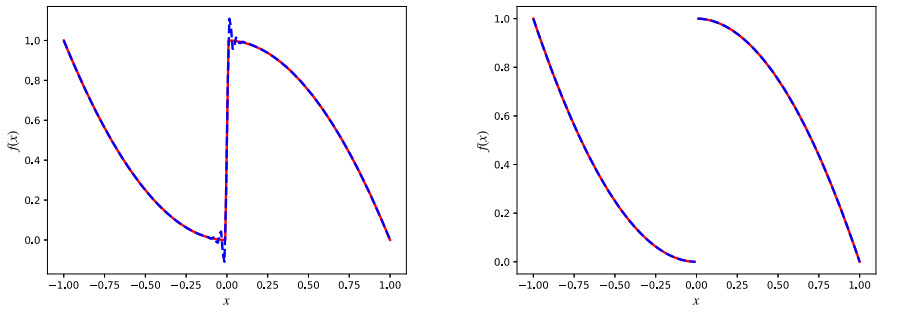
By inserting the constraints (11) into (17), we then have

$$\begin{aligned} &\sum_{i=1}^{n} \lambda_i \mathcal{D}(\phi)(\|\boldsymbol{x}_1 - \boldsymbol{x}_i\|) + \sum_{j=1}^{n} \alpha_j \mathcal{D}(\psi_j)(\boldsymbol{x}_1) \\ &\quad = \sum_{i=1}^{n} \lambda_i \sum_{k=1}^{n} w_k \phi_{ik} + \sum_{j=1}^{m} w_{n+j} \sum_{i=1}^{n} \lambda_i \psi_{ij} + \sum_{j=1}^{n} \alpha_j \sum_{k=1}^{n} w_k \psi_{kj} \\ &\quad = \sum_{i=1}^{n} \lambda_i \left(\sum_{k=1}^{n} w_k \phi_{ik} + \sum_{j=1}^{m} w_{n+j} \psi_{ij} \right) + \sum_{j=1}^{n} \alpha_j \sum_{k=1}^{n} w_k \psi_{kj}. \end{aligned} \tag{18}$$

due to the fact $\sum_{i=1}^{n} \lambda_i \psi_{ij} = 0$ for all $1 \le j \le m$. Comparing the left and right sides of (18) yields:

$$\begin{pmatrix} \phi_{11} & \cdots & \phi_{1n} & \psi_{11} & \cdots & \psi_{1m} \\ \vdots & & \vdots & \vdots & & \vdots \\ \phi_{n1} & \cdots & \phi_{nn} & \psi_{n1} & \cdots & \psi_{nm} \\ \psi_{11} & \cdots & \psi_{n1} & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & & \vdots \\ \psi_{1m} & \cdots & \psi_{nm} & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} w_1 \\ \vdots \\ w_n \\ w_{n+1} \\ \vdots \\ w_{n+m} \end{pmatrix} = \begin{pmatrix} \mathcal{D}(\phi)(r_{11}) \\ \vdots \\ \mathcal{D}(\phi)(r_{1n}) \\ \mathcal{D}(\psi_1)(\boldsymbol{x}_1) \\ \vdots \\ \mathcal{D}(\psi_m)(\boldsymbol{x}_1) \end{pmatrix} \tag{19}$$

Then, one solves this above symmetric linear system, and the resulting $(w_1, \ldots, w_n)$ are the coefficients we seek, while $(w_{n+1}, \ldots, w_{n+m})$ are simply ignored. Utilizing the condition



(a) Standard RBF interpolation (b) A RBF interpolation with domain decomposition

Fig. 2 Numerical results of two types of RBF interpolation for discontinuous function.

$\sum_{i=1}^{n} w_i = 0$ (obtained by using $\psi_1 \equiv 1$), (16) can be further rewritten as:

$$\mathcal{D}(u)(\boldsymbol{x}_1) \approx \sum_{i=1}^{n} w_i u_i - \sum_{i=1}^{n} w_i u_1 = \sum_{i=2}^{n} w_i (u_i - u_1), \tag{20}$$

which highlights why this is referred to as a (generalized) *finite difference approximation*.

3 The Proposed RBF-FD Method for Linear Anisotropic Elliptic Interface Problems

In this section, we will illustrate the proposed RBF-FD method for solving linear the anisotropic elliptic interface problem (1), together with the motivations behind the method.

3.1 Exploring the Challenge of RBF Interpolation for Discontinuous Functions

Before delving into the details of our RBF-FD method, let us first explore the challenge associated with RBF interpolation for discontinuous functions through an heuristic example. Consider the following function in 1D:

$$f(x) = \begin{cases} x^2, & -1 < x \leq 0, \\ 1 - x^2, & 0 < x < 1, \end{cases}$$

which has a discontinuity at $x = 0$. We approach the interpolation of this function using the RBF interpolation method in two distinct ways.

The first approach applies standard RBF interpolation directly over the entire domain. While this method provides accurate interpolation away from the discontinuity, it often results in significant oscillations near the discontinuity. This is due to the inability of standard RBFs to effectively handle abrupt changes, as demonstrated in Figure 2a . The second approach utilizes domain decomposition, dividing the domain into two sub-domains based on the location of the discontinuity. RBF interpolation is then applied independently in each sub-domain, ensuring that the scattered points used to construct the interpolation basis functions do not cross the discontinuity. This method helps avoid oscillations near the discontinuity, as each sub-domain is treated as a continuous region. It is especially effective when the location of the discontinuity is known, providing more stable and accurate interpolation, as illustrated

in Figure 2b . This example motivates the main idea of our RBF-FD method for solving linear elliptic interface problems, as such problems typically involve solutions with inherent discontinuities.

3.2 Domain Decomposition and the RBF-FD Discretization Scheme

In the following, we will elaborate on our RBF-FD method that utilizes domain decomposition to effectively handle the linear anisotropic elliptic interface problem (1). For convenience, we will describe our proposed method through a 2D example with two subdomains Ω^+ (the interior one) and Ω^- (the exterior one) separated by a closed interface Γ, which does not affect the completeness of the method.

Domain Decomposition-Based Point Sets

Since RBF-FD is a mesh-free method, points must be scattered across the entire computational domain $\overline{\Omega}$, including the interface and boundary. These points form a point set $\mathcal{P} = \{\boldsymbol{x}_i\}_{i=1}^{M}$, where M is the total number of points. The points are then divided into four distinct subsets:

$$\mathcal{P} = \mathcal{P}_{\Omega^+} \cup \mathcal{P}_{\Omega^-} \cup \mathcal{P}_{\partial} \cup \mathcal{P}_{\Gamma},$$

corresponding to the scattered points in the subdomains Ω^+ andΩ^- and on the the boundary $\partial\Omega$ and the interface Γ (see Figure 3a) respectively. These points will be assigned global indices. The global index sets for $\mathcal{P}_{\Omega^+}$, $\mathcal{P}_{\Omega^-}$ and $\mathcal{P}_{\partial}$ are denoted by Θ_{Ω^+}, Θ_{Ω^-} and Θ_{∂}, respectively. Due to the dual usage of all points on the interface in the proposed RBF-FD method, each interface point is assigned *two global indices*. These indices determine two index sets for the interface points $\mathcal{P}_{\Gamma}$, namely Θ_{Γ^+} and Θ_{Γ^-}. All of these index sets together form the complete index set, denoted as $\Theta = \{i\}_{i=1}^{N}$. Clearly, the relationship $N - M = |\mathcal{P}_{\Gamma}|$ holds, where $|\cdot|$ denotes the number of elements in the set.

Additionally, for the convenience of the algorithm construction, we define the following index sets, one for each of the two subdomains:

$$\Theta^+ = \Theta_{\Omega^+} \cup \Theta_{\Gamma^+} \cup \Theta_{\partial}$$

for the subdomain Ω^+ and

$$\Theta^- = \Theta_{\Omega^-} \cup \Theta_{\Gamma^-} \cup \Theta_{\partial}.$$

for the subdomain Ω^-, respectively.

The RBF-FD approximation of the governing equation in the two subdomains

Since domain decomposition is employed, we focus on illustration of the RBF-FD approximations for points in Ω^+, with an analogous process applied in Ω^-. Assume i_1 be an arbitrary global index in Θ_{Ω^+} (i.e., $\boldsymbol{x}_{i_1} \in \Omega^+$) and $\Xi_{i_1} = \{i_1, i_2, \cdots, i_n\} \subset \Theta^+$ be the set of global indices of the n nearest neighbors of $\boldsymbol{x}_{i_1}$ belonging to $\overline{\Omega^+} = \Omega^+ \cup \Gamma$ (i.e., these neighbors are not allowed to cross the interface Γ to reach Ω^-), including itself. Ξ_{i_1} is commonly referred to as a *stencil* of the index i_1 (see Figure 3b). Based on the analysis in Section 2.2, the RBF-FD scheme for approximating the governing equation (1a) at $\boldsymbol{x}_{i_1}$ using this stencil can be formulated as follows:

$$\sum_{k=1}^{n} w_{i_k}^{*} u_{i_k} = f(\boldsymbol{x}_{i_1}) - \sigma(\boldsymbol{x}_{i_1}) u_{i_1}, \tag{21}$$

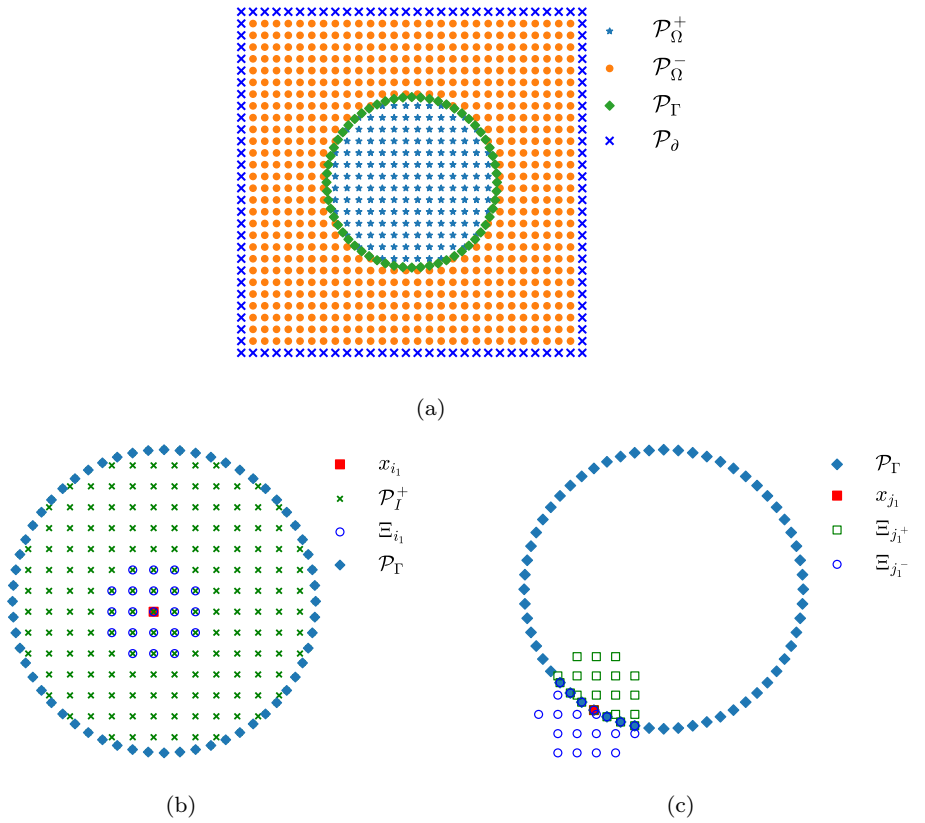


Fig. 3 (a) Different types of point sets based on domain decomposition, (b) the stencil used for discretizing the governing equation at the point $\boldsymbol{x}_{i_1}$ in the subdomain Ω^+, and (c) the two stencils used for approximating the jump condition of differential type at the point $\boldsymbol{x}_{j_1}$ on the interface Γ.

where the coefficients $(w_{i_1}^*, \cdots, w_{i_n}^*)$ are determined by solving the following small linear system

$$\begin{pmatrix} \phi_{i_1,i_1} & \cdots & \phi_{i_1,i_n} & \psi_{i_1,1} & \cdots & \psi_{i_1,m} \\ \vdots & & \vdots & \vdots & & \vdots \\ \phi_{i_n,i_1} & \cdots & \phi_{i_n,i_n} & \psi_{i_n,1} & \cdots & \psi_{i_n,m} \\ \psi_{i_1,1} & \cdots & \psi_{i_n,1} & 0 & \cdots & 0 \\ \vdots & & \vdots & 0 & \cdots & 0 \\ \psi_{i_1,m} & \cdots & \psi_{i_n,m} & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} w_{i_1}^* \\ \vdots \\ w_{i_n}^* \\ w_{i_{n+1}}^* \\ \vdots \\ w_{i_{n+1m}}^* \end{pmatrix} = \begin{pmatrix} \mathcal{D}(\phi)(r_{i_1,i_1}) \\ \vdots \\ \mathcal{D}(\phi)(r_{i_1,i_n}) \\ \mathcal{D}(\psi_1)(\boldsymbol{x}_{i_1}) \\ \vdots \\ \mathcal{D}(\psi_m)(\boldsymbol{x}_{i_1}) \end{pmatrix}. \tag{22}$$

where ϕ_{i_k,i_l} $(i_k, i_l \in \Xi_{i_1})$ and $\psi_{i_k,l}$ $(i_k \in \Xi_{i_1}, l = 1, \cdots, m)$ are defined by (9) and (14), respectively, and

$$\mathcal{D}(\phi) := \mathcal{L}(u) - \sigma u = -\nabla \cdot (\Lambda \nabla \phi).$$

The RBF-FD approximation of the jumping conditions on the interface For elliptic interface problems, the discretization of jump conditions on the interfaces is also very crucial. For any point $\boldsymbol{x}_{j_1}$ on the interface Γ, it has two global indices, $j_1^+ \in \Theta_{\Gamma^+}$ (associated with the

subdomain Ω^+) and $j_1^- \in \Theta_{\Gamma^-}$ (associated with the subdomain Ω^-). For the second-type jump condition (i.e., function value type) (1b), the treatment is straightforward, i.e.,

$$u_{j_1^+} - u_{j_1^-} = h(\boldsymbol{x}_{j_1}). \tag{23}$$

To discretize the second-type jump condition (i.e., normal differential type) (1c), we will introduce two stencils for use, one from each of two subdomains:

$$\Xi_{j_1^+} = \{j_1^+, j_2^+, \cdots, j_n^+\} \subset \Theta^+ \quad \text{corresponding to} \quad j_1^+ \in \Theta_{\Gamma^+},$$

which is the set of global indices of the n nearest neighbors of $\boldsymbol{x}_{j_1}$ belonging to $\overline{\Omega}^+$ and

$$\Xi_{j_1^-} = \{j_1^-, j_2^-, \cdots, j_n^-\} \subset \Theta^- \quad \text{corresponding to} \quad j_1^- \in \Theta_{\Gamma^-},$$

which is the set of global indices of the n nearest neighbors of $\boldsymbol{x}_{j_1}$ belonging to $\overline{\Omega^-}$ (see Figure 3c). Then, the jump condition (1c) at the point $\boldsymbol{x}_{j_1}$ can be approximated by

$$\sum_{k=1}^{n} w_{j_k^+}^* u_{j_k^+} - \sum_{k=1}^{n} w_{j_k^-}^* u_{j_k^-} = q(\boldsymbol{x}_{j_1}), \tag{24}$$

where $\{w_{j_k^+}^*\}_{k=1}^n$ and $\{w_{j_k^-}^*\}_{k=1}^n$ are determined by solving the following small linear systems

$$\begin{pmatrix} \phi_{j_1^+,j_1^+} & \cdots & \phi_{j_1^+,j_n^+} & \psi_{j_1^+,1} & \cdots & \psi_{j_1^+,m} \\ \vdots & & \vdots & \vdots & & \vdots \\ \phi_{j_n^+,j_1^+} & \cdots & \phi_{j_n^+,j_n^+} & \psi_{j_n^+,1} & \cdots & \psi_{j_n^+,m} \\ \psi_{j_1^+,1} & \cdots & \psi_{j_n^+,1} & 0 & \cdots & 0 \\ \vdots & & \vdots & 0 & \cdots & 0 \\ \psi_{j_1^+,m} & \cdots & \psi_{j_n^+,m} & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} w_{j_1^+} \\ \vdots \\ w_{j_n^+} \\ w_{j_{n+1}^+} \\ \vdots \\ w_{j_{n+m}^+} \end{pmatrix} = \begin{pmatrix} \mathcal{J}(\phi)(r_{j_1^+,j_1^+}) \cdot \boldsymbol{n} \\ \vdots \\ \mathcal{J}(\phi)(r_{j_1^-,j_n^-}) \cdot \boldsymbol{n} \\ \mathcal{J}(\psi_1)(\boldsymbol{x}_{j_1^+}) \cdot \boldsymbol{n} \\ \vdots \\ \mathcal{J}(\psi_m)(\boldsymbol{x}_{j_1^+}) \cdot \boldsymbol{n} \end{pmatrix} \tag{25}$$

and

$$\begin{pmatrix} \phi_{j_1^-,j_1^-} & \cdots & \phi_{j_1^-,j_n^-} & \psi_{j_1^-,1} & \cdots & \psi_{j_1^-,m} \\ \vdots & & \vdots & \vdots & & \vdots \\ \phi_{j_n^-,j_1^-} & \cdots & \phi_{j_n^-,j_n^-} & \psi_{j_n^-,1} & \cdots & \psi_{j_n^-,m} \\ \psi_{j_1^-,1} & \cdots & \psi_{j_n^-,1} & 0 & \cdots & 0 \\ \vdots & & \vdots & 0 & \cdots & 0 \\ \psi_{j_1^-,m} & \cdots & \psi_{j_n^-,m} & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} w_{j_1^-} \\ \vdots \\ w_{j_n^-} \\ w_{j_{n+1}^-} \\ \vdots \\ w_{j_{n+m}^-} \end{pmatrix} = \begin{pmatrix} \mathcal{J}(\phi)(r_{j_1^-,j_1^-}) \cdot \boldsymbol{n} \\ \vdots \\ \mathcal{J}(\phi)(r_{j_1^-,j_n^-}) \cdot \boldsymbol{n} \\ \mathcal{J}(\psi_1)(\boldsymbol{x}_{j_1^-}) \cdot \boldsymbol{n} \\ \vdots \\ \mathcal{J}(\psi_m)(\boldsymbol{x}_{j_1^-}) \cdot \boldsymbol{n} \end{pmatrix}, \tag{26}$$

respectively, while $\phi_{j_k^\pm,j_l^\pm}(j_k^\pm, j_l^\pm \in \Xi_{j_1^\pm})$ and $\psi_{j_k^\pm,l}(j_k^\pm \in \Xi_{j_1^\pm},\ l = 1, \cdots, m)$ are defined by (9) and (14), respectively.

Boundary treatment For the Dirichlet boundary condition (1d), we simply set

$$u_i = g(\boldsymbol{x}_i), \tag{27}$$

for all $i \in \Theta_\partial$.

3.3 Implementation Algorithm and Discussions

The complete RBF-FD method, for solving the linear anisotropic elliptic interface problem (1), with two subdomains Ω^+ and Ω^- separated by a single interface Γ, is elaborated in Algorithm 1. Note that construction of the RBF-FD approximations of functions and differential operators in each of the subdomains independently is essential in the proposed method in order to resolve inherent discontinuities on the interfaces. The generalization of Algorithm 1 to the problems of multiple interfaces and those of three dimensions is straightforward and thus the description is skipped here.

Algorithm 1 The Proposed RBF-FD Method

Require: The domain Ω, the interface Γ, the diffusion tensor Λ, the reaction coefficients σ, the boundary value g, the jump conditions h and q, the stencil size n and the order p of the polynomial basis functions, and the set of points $\mathcal{P}$

Form the global index sets Θ_{Ω^+}, Θ_{Ω^-}, Θ_{Γ^+}, Θ_{Γ^+} and Θ_∂ for the points $\mathcal{P}$ based on the domain Ω and the interface Γ

Initialize the stiffness matrix $\boldsymbol{A} = (a_{ij}) \in \mathbb{R}^{N\times N}$ and the right-hand side vector $\boldsymbol{b} = (b_i) \in \mathbb{R}^N$

for $i_1 \in \Theta_{\Omega^+} \cup \Theta_{\Omega^-}$ **do** ▷ Discretize the governing equation at interior points of subdomains

  Solve the linear system (22) to obtain $\left\{w^*_{i_k}\right\}_{k=1}^n$ in (21)

  Assemble $\boldsymbol{A}$ by setting $a_{i_1,i_k} = w^*_{i_k} + \delta_{i_1 i_k}\sigma(\boldsymbol{x}_{i_1})$ for $k = 1, \cdots, n$

  Update $\boldsymbol{b}$ by setting $b_{i_1} = f(\boldsymbol{x}_{i_1})$

end for

for $j_1^+ \in \Theta_{\Gamma^+}$ **do** ▷ Treat the first jump condition at interface points

  Assemble $\boldsymbol{A}$ by setting $a_{j_1^+,j_1^+} = 1$ and $a_{j_1^+,j_1^-} = -1$

  Update $\boldsymbol{b}$ by setting $b_{j_1^+} = h(\boldsymbol{x}_{j_1})$

end for

for $j_1^+ \in \Theta_{\Gamma^+}$ **do** ▷ Treat the second jump condition at interface points

  Solve the linear systems (25) and (25) to obtain $\left\{w^*_{j_k^+}\right\}_{k=1}^n$ and $\left\{w^*_{j_k^-}\right\}_{k=1}^n$ in (24)

  Assemble $\boldsymbol{A}$ by setting $a_{j_1^-,j_k^+} = w_{j_k^+}$ and $a_{j_1^-,j_k^-} = -w_{j_k^-}$ for $k = 1, \cdots, n$

  Update $\boldsymbol{b}$ by setting $b_{j_1^-} = q(\boldsymbol{x}_{j_1})$

end for

for $i \in \Theta_\partial$ **do** ▷ Treat the Dirichlet boundary condition at boundary points

  Assemble $\boldsymbol{A}$ by setting $a_{i,i} = 1$

  Update $\boldsymbol{b}$ by setting $b_i = g(\boldsymbol{x}_i)$

end for

Solve the linear system $\boldsymbol{Au} = \boldsymbol{b}$ using a sparse linear system solver

Return: The numerical solution $\boldsymbol{u}$

We can use a KD-tree to store the nodes, resulting in a stencil construction cost of $\mathcal{O}(N \log N)$. It is clear that the resulting stiffness matrix of the RBF-FD method is highly sparse, and as the stencil size increases, the number of the non-zero elements also increases. We can make use of popular sparse linear system solvers for the solution process. In addition, a small linear system of $(n+m)$ variables needs to be solved for each node before assembling

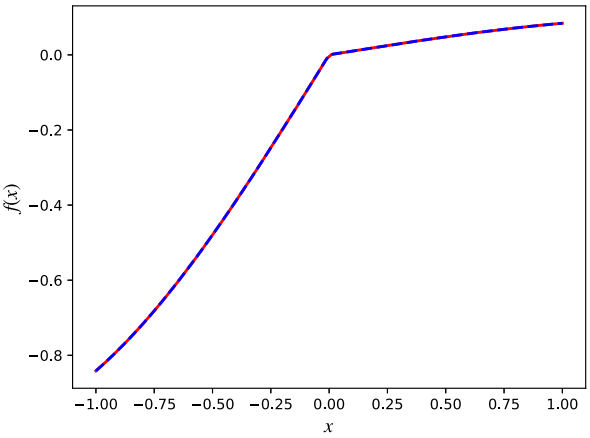


Fig. 4 The standard RBF interpolation for a sample continuous but non-smooth function.

the global stiffness matrix, the computational complexity is $\mathcal{O}((n+m)^3 N)$ for the this step. Consequently, the proposed high-order RBF-FD method has a potential advantage over the meshless methods like ELM [19] and RFM [4] developed in recent years for solving elliptic interface problems – those methods eventually reduce to solving a least-squares problem, for which sparse solvers are generally lacking and such limitation may restrict them from solving very larger-scale problems.

The proposed RBF-FD method also can be directly applied to the classic anisotropic diffusion equation with discontinuous diffusion tensors, described as follows:

$$\begin{cases} -\nabla \cdot (\Lambda \nabla u) = f, & \boldsymbol{x} \in \Omega, \\ u = g, & \boldsymbol{x} \in \partial\Omega, \end{cases} \tag{28}$$

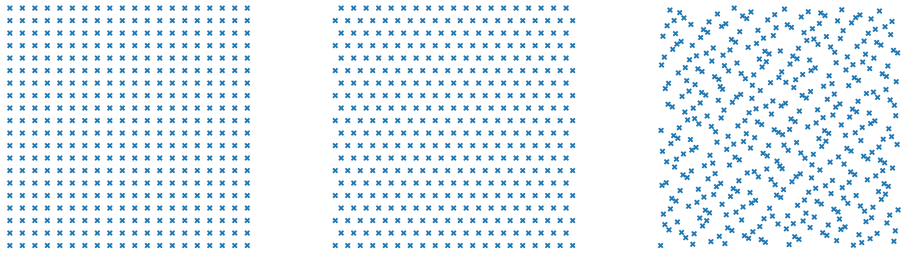
It is important to note that the diffusion equation (28) does not explicitly include jump conditions on the interface Γ where the diffusion tensor is discontinuous. However, the following constraints can be derived from physical principles:

$$\begin{cases} [\![u]\!] = 0, & \boldsymbol{x} \in \Gamma, \\ [\![\Lambda \nabla u \cdot \boldsymbol{n}]\!] = 0, & \boldsymbol{x} \in \Gamma, \end{cases} \tag{29}$$

This is equivalent to setting h and q in (1) to zero. Additionally, it should be emphasized that in this context, stencil nodes could be allowed to cross the interface without introducing oscillations. This is because the solution of the problem (28) remains continuous, even though its gradient exhibits discontinuities. In this case the function still can be well approximated by RBF interpolation, see Figure 4 for an illustration.

4 Numerical Experiments

In this section, we present numerical experiments in two and three dimensions to demonstrate the high-order accuracy and computational efficiency of the proposed RBF-FD method for solving various linear anisotropic elliptic interface problems taking the form (1). The choice of method for generating scatter points is often important in determining the accuracy of numerical simulations, particularly when mesh-free methods like RBF-FD are employed. To describe the characteristics of a given scattered point set $\mathcal{P}$, we introduce several metrics,



(a) Cartesian (b) Hexagonal (c) Halton

Fig. 5 Three types of points layouts for the RBF-FD method.

including the fill distance h, the separation q, and the mesh ratio ρ, defined as follows [12]:

$$h = \sup_{\boldsymbol{x}\in\Omega} \inf_{\boldsymbol{x}_i\in\mathcal{P}} \|\boldsymbol{x} - \boldsymbol{x}_i\|, \qquad q = \frac{1}{2} \inf_{\boldsymbol{x}_i,\boldsymbol{x}_j\in\Omega, i\neq j} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|, \qquad \rho = \frac{h}{q}.$$

Clearly, ρ is always greater than or equal to 1, and the closer ρ is to 1, the more uniformly distributed the point set $\mathcal{P}$ is. As discussed in [12], we prefer point sets where ρ is close to 1, indicating a more uniform distribution. There are three commonly used point layout schemes: Cartesian, Hexagonal, and Halton. Figure 5 illustrates the scatter points generated by these three layouts. It is evident that the mesh ratio ρ of the third point set is greater than that of the first two. Through comparisons in our numerical experiments, the results show that the RBF-FD method performs better with the first two layouts compared to the third, with similar performance between the first two. Similar observations were also found in [12]. Thus, we only report the results on the Cartesian scattered points for simplicity in this paper.

We first consider four different types of interface settings in 2D: 1) a line interface, 2) a circular interface, 3) a flower-shaped interface, and 4) a case of three mixed interfaces in Example 1-4, see Figure 6 for illustration. Example 5 tests a 3D problem of two spherical surface interfaces (analogous to circular interfaces in 2D). In all examples, we apply the RBF-FD method with four level of node sets. For each point set level k ($k = 1, 2, 3, 4$), the number of total nodes is set to be $N = (N_k)^d$. We also set $|\Theta_\Gamma| = N_k$ in Example 1, $|\Theta_\Gamma| = 2N_k$ in Example 2, $|\Theta_\Gamma| = 3N_k$ in Example 3 and $\Theta_\Gamma = \Theta_{\Gamma_1} \cup \Theta_{\Gamma_2} \cup \Theta_{\Gamma_3}$ with $|\Theta_{\Gamma_1}| = N_k$, $|\Theta_{\Gamma_2}| = 2N_k$, $|\Theta_{\Gamma_3}| = 3N_k$ in Example 4, and $\Theta_\Gamma = \Theta_{\Gamma_1} \cup \Theta_{\Gamma_2}$ with $|\Theta_{\Gamma_1}| = |\Theta_{\Gamma_2}| = N_k^2$ in Example 5.

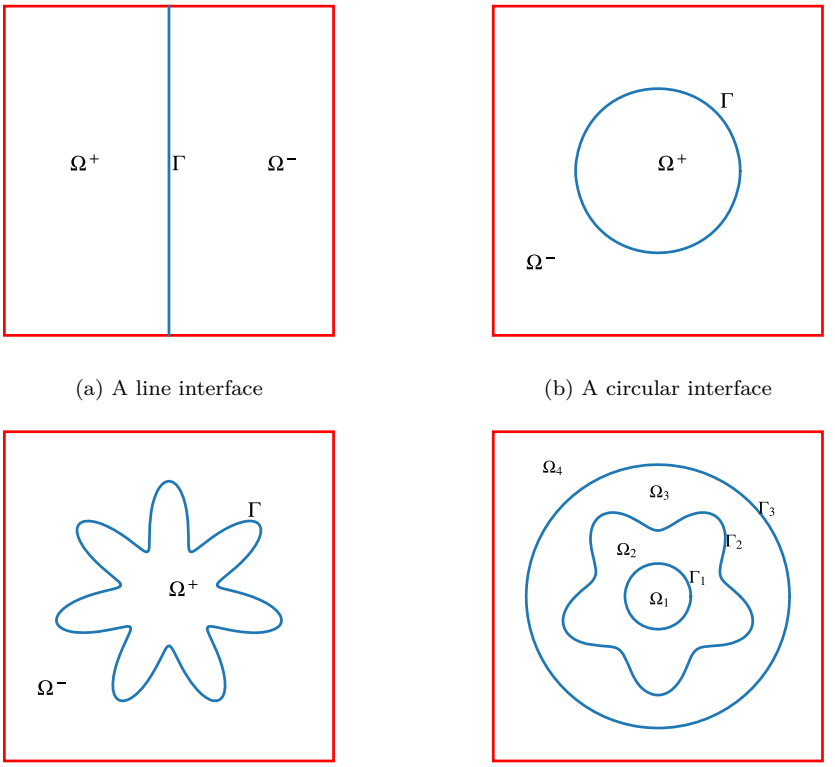
To evaluate the accuracy, we introduce the relative L_2 error:

$$err_{L2} = \sqrt{\frac{\sum_{i=1}^{N} |u_i - u_i^*|^2}{\sum_{i=1}^{N} |u_i|^2}}$$

where u_i denotes the numerical solution obtained with the RBF-FD method, while u^* represents the exact solution. Then we quantify the order of convergence of the proposed RBF-FD method as follow

$$\frac{\log(err_{L2}(N_k)/err_{L2}(N_{k-1}))}{\log(N_{k-1}/N_k)}$$

where $err_{L2}(N_k)$ is the corresponding relative L_2 error. All tests were conducted on a Windows system equipped with an Intel Core i5-13600KF processor.



(a) A line interface

(b) A circular interface

(c) A flower-shaped interface

(d) A case of three mixed interfaces

Fig. 6 Four types of sample interface settings for numerical tests in 2D.

4.1 Example 1: A 2D anisotropic diffusion equation with a line interface

Firstly, we consider a 2D anisotropic diffusion equation (28) in the domain $\Omega = (0, 1) \times (0, 1)$, with a line interface

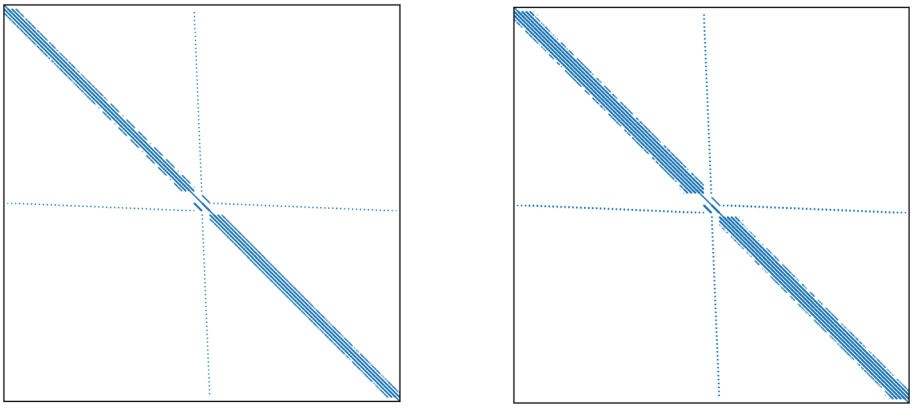
$$\Gamma = \{(x, y) \mid x = 0.5,\ 0 < y < 1\}$$

dividing Ω into two parts, Ω^+ and Ω^-, see Figure 6a . The high contrast diffusion tensor and the exact solution are chosen as follows:

$$\Lambda(x, y) = \begin{cases} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & (x, y) \in \Omega^+, \\ \begin{pmatrix} 10^3 & 0 \\ 0 & 10^{-2} \end{pmatrix}, & (x, y) \in \Omega^-, \end{cases} \tag{30}$$

and

$$u(x, y) = \begin{cases} \cos(\pi x)\sin(\pi y), & (x, y) \in \Omega^+, \\ 10^{-3}\cos(\pi x)\sin(\pi y), & (x, y) \in \Omega^-. \end{cases} \tag{31}$$



(a) $(n, p) = (12, 2)$ (b) $(n, p) = (25, 4)$

Fig. 7 The distribution of non-zero elements in the stiffness matrix $\boldsymbol{A}$ of Example 1.

The source term f and boundary conditions g are determined accordingly. In this example, the jump conditions are automatically homogeneous, i.e., $h = q = 0$.

We apply the RBF-FD method with two different (n, p) settings on scattered points at different levels to solve this problem. In Figure 7, we show the structure of the global stiffness matrix for the RBF-FD method under two different configurations, $(n, p) = (12, 2)$ and $(25, 4)$. Note that $m = 6$ when $p = 2$ and $m = 15$ when $p = 4$ for 2D problems, thus the invertibility condition $n \geq m$ for the local linear system is satisfied for both two configurations. The corresponding numerical results and running times are reported in Table 2. It can be observed that the average convergence rates for the two configurations are 2 and 4, respectively, which matches our expectations. Additionally, the running times for both configurations exhibit an approximately linear scaling with respect to the total number of scattered points (i.e., N) and the number of stencil points (i.e., n). For this example, due to the continuity of the exact solutions and the high-order accuracy of the schemes, a relatively small number of scattered points can achieve very small errors, for instance, the errors quickly reduce to the level of $\mathcal{O}(10^{-7})$ with 48^2 points for the 4th-order scheme (i.e., $p = 4$). In addition, Figure 8 provides the heat map and error map of the approximate solution obtained by the RBF-FD method with $(n, p) = (25, 4)$ on the densest set of points ($N = 48^2$). It can be seen that the errors in the solution region are very small, with slightly larger errors appearing at the interface or boundary.

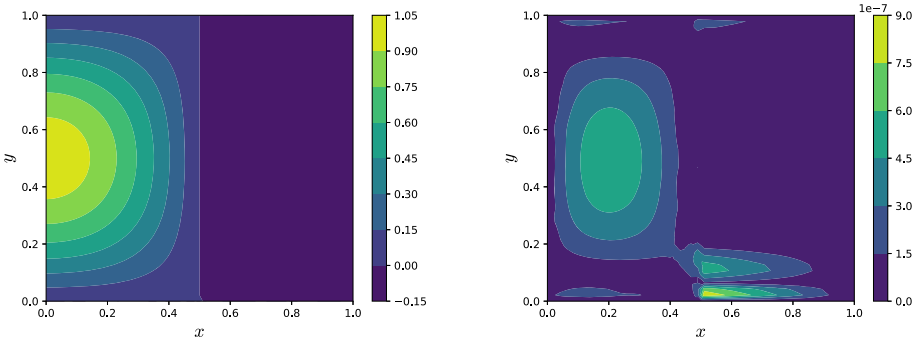
4.2 Example 2: A 2D Anisotropic Diffusion Interface Problem with a Circular Interface

Next, we consider a 2D anisotropic diffusion interface problem (1) (i.e., without reaction) in the domain $\Omega = (-1, 1) \times (-1, 1)$, with a circular interface

$$\Gamma = \{(x, y) \mid x = 0.5 \cos\theta,\ y = 0.5 \sin\theta,\ \theta \in [0, 2\pi)\}$$

Table 2 Numerical results and running times (measured in seconds) of the proposed RBF-FD method for solving the 2D anisotropic diffusion problem in Example 1.

	$(n, p) = (12, 2)$				$(n, p) = (25, 4)$			
N	err_{L2}	Order	Sparsity(%)	Time(s)	err_{L2}	Order	Sparsity(%)	Times(s)
12^2	3.08e-3		5.34	0.012	7.47e-4		10.65	0.021
24^2	8.61e-4	1.84	1.84	0.047	9.81e-6	6.25	3.51	0.086
36^2	3.77e-4	2.03	0.80	0.107	1.63e-6	4.42	1.69	0.196
48^2	2.28e-4	1.73	0.46	0.191	5.87e-7	3.55	0.99	0.361
Ave. Ord.		**1.87**				**4.74**		



(a) Heat map (b) Error map

Fig. 8 Plots of the heat map (left) and point-wise error map (right) of numerical solution produced by the proposed RBF-FD method with $(n, p) = (25, 4)$ on the densest set of points ($N = 48^2$) in Example 1.

as illustrated in Figure 6b . We choose two types of diffusion tensors, one being piecewise constant and the other being a piecewise function. Specifically, we define them as:

$$(a) : \Lambda(x, y) = \begin{cases} \begin{pmatrix} 1 & 0 \\ 0 & 0.1 \end{pmatrix}, & (x, y) \in \Omega^+, \\ \begin{pmatrix} 1 & 0 \\ 0 & \beta \end{pmatrix}, & (x, y) \in \Omega^-, \end{cases} \tag{32}$$

$$(b) : \Lambda(x, y) = \begin{cases} \begin{pmatrix} x^2 + 3 & \cos y \\ \cos y & y^2 + 3 \end{pmatrix}, & (x, y) \in \Omega^+, \\ \beta \begin{pmatrix} x^2 + 3 & \cos y \\ \cos y & y^2 + 3 \end{pmatrix}, & (x, y) \in \Omega^-, \end{cases} \tag{33}$$

where $\beta = 10^3$. The corresponding exact solutions are chosen as follows:

$$(a) : u(x, y) = \begin{cases} \cos(1 - x^2 + y^2), & (x, y) \in \Omega^+, \\ \sin(x^2 + y^2), & (x, y) \in \Omega^-, \end{cases} \tag{34}$$

$$(b) : u(x, y) = \begin{cases} \sin(10xy), & (x, y) \in \Omega^+, \\ \sin(5xy) + 1, & (x, y) \in \Omega^-. \end{cases} \tag{35}$$

The source terms f, boundary conditions g and jump conditions h, φ are determined accordingly.

We use the RBF-FD method with two configurations $(n, p) = (12, 2)$ and $(25, 4)$ for most points, except those very close to the interface or boundary, or on the interface, where $(n, p) = (24, 2)$ and $(40, 4)$ are used, respectively. The corresponding numerical results and running times are presented in Table 3. For the two different types of diffusion tensors, the performance of RBF-FD is similar, and both achieve the expected convergence accuracy, demonstrating the broad applicability of RBF-FD in handling interface problems. We again observe that the running times for both configurations exhibit an approximately linear scaling with respect to the total number of scattered points (i.e., N) and the number of stencil points (i.e., n). Although the exact solutions have discontinuities, the errors still quickly reduce to the level of $\mathcal{O}(10^{-6})$ with 120^2 points for the 4th-order scheme. Comparing the results on Tables 2 and 3, it can be seen that solving anisotropic interface problems clearly requires more scattered points than anisotropic diffusion problems to achieve the same level of accuracy using the RBF-FD method. Additionally, we also show some qualitative results of the RBF method with $(n, p) = (25, 4)$ on the densest set of points ($N = 120^2$) in Figures 9, such as surface plots, heat maps, and error maps of the approximate solutions. From the error maps, it can be observed that the errors in the domain are generally very small, with slightly larger errors concentrated near the interface.

4.3 Example 3: A 2D Anisotropic Elliptic Interface Problem with a Flower-Shape Interface

We consider a more challenging anisotropic elliptic interface problem (1) in the domain $\Omega = (-1, 1) \times (-1, 1)$, where the interface is more complex Γ and flower-shaped, defined by

$$\begin{cases} x = (0.5 - 0.2\sin(7\theta))\cos(\theta), \\ y = (0.5 - 0.2\sin(7\theta))\sin(\theta), \end{cases}$$

where $\theta \in [0, 2\pi)$, as illustrated in Figure 6c . In this case, we consider two types of diffusion tensors and reaction coefficients, both of which are discontinuous across the interface, defined as follows:

$$\text{(a)}: \quad \mathcal{A}(x, y) = \begin{cases} \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}, & (x, y) \in \Omega^+, \\ \begin{pmatrix} 40 & 30 \\ 30 & 60 \end{pmatrix}, & (x, y) \in \Omega^-, \end{cases} \tag{36}$$

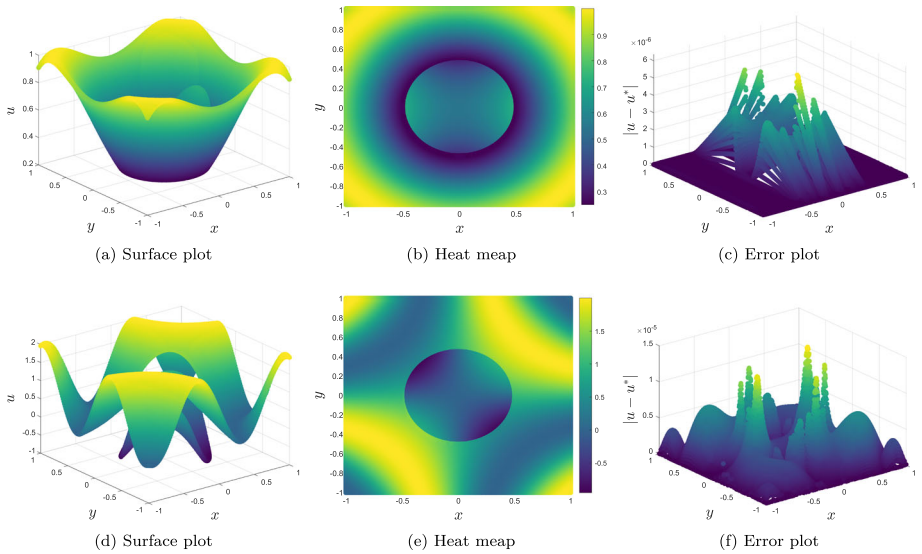
$$\sigma(x, y) = \begin{cases} 1, & (x, y) \in \Omega^+, \\ 10, & (x, y) \in \Omega^-, \end{cases} \tag{37}$$

$$\text{(b)}: \quad \mathcal{A}(x, y) = \begin{cases} \begin{pmatrix} x^2 + y^2 + 1 & x^2 + y^2 \\ x^2 + y^2 & x^2 + y^2 + 2 \end{pmatrix}, & (x, y) \in \Omega^+, \\ \beta \begin{pmatrix} x^2 + y^2 + 1 & x^2 + y^2 \\ x^2 + y^2 & x^2 + y^2 + 2 \end{pmatrix}, & (x, y) \in \Omega^-, \end{cases} \tag{38}$$

$$\sigma(x, y) = \begin{cases} \exp(x)(x^2 + y^2 + 1)\cos(y), & (x, y) \in \Omega^+, \\ \beta \exp(x)(x^2 + y^2 + 1)\cos(y), & (x, y) \in \Omega^-, \end{cases} \tag{39}$$

Table 3 Numerical results and running times (measured in seconds) of the proposed RBF-FD method for solving the 2D anisotropic diffusion interface problem in Example 2.

	Example 2(a)						Example 2(b)					
	$(n, p) = (12, 2)$			$(n, p) = (25, 4)$			$(n, p) = (12, 2)$			$(n, p) = (25, 4)$		
N	err_{L2}	Order	Time(s)	err_{L2}	Order	Time(s)	err_{L2}	Order	Time(s)	err_{L2}	Order	Time(s)
15^2	8.13e-2		0.027	3.79e-2		0.048	2.43e-1		0.029	2.25e-1		0.049
30^2	1.80e-2	2.17	0.101	5.24e-3	2.84	0.180	6.48e-2	1.90	0.105	6.07e-3	5.21	0.194
60^2	7.10e-3	1.34	0.389	3.79e-4	3.79	0.706	7.11e-3	3.18	0.426	2.79e-4	4.43	0.748
120^2	1.03e-3	2.77	1.553	5.61e-6	6.08	2.944	2.21e-3	1.68	1.629	7.09e-6	5.30	3.033
Ave. Ord.		**2.09**			**4.24**			**2.25**			**4.98**	



(a) Surface plot (b) Heat meap (c) Error plot

(d) Surface plot (e) Heat meap (f) Error plot

Fig. 9 Plots of the surface, heat map, and point-wise error maps of numerical solutions by the proposed RBF-FD method with $(n, p) = (25, 4)$ on the densest set of points ($N = 120^2$) in Example 2. Top row: Example 2(a); bottom row: Example 2(b).

where $\beta = 10^3$. The corresponding exact solutions are given by

$$\text{(a)}: \quad u(x, y) = \begin{cases} \cos(2xy) + y, & (x, y) \in \Omega^+, \\ \sin(2xy) + x, & (x, y) \in \Omega^-, \end{cases} \tag{40}$$

$$\text{(b)}: \quad u(x, y) = \begin{cases} \exp(x)\cos(y), & (x, y) \in \Omega^+, \\ \log(y + 2) - x, & (x, y) \in \Omega^-. \end{cases} \tag{41}$$

The source terms f, boundary conditions g and jump conditions h, φ are determined accordingly.

The corresponding results and running times of the RBF-FD method with the two configurations same as Example 2 are presented in Table 4 and Figures 10. The numerical performance is similar to that of the previous example, demonstrating the applicability of the RBF-FD method in handling complex interface problems.

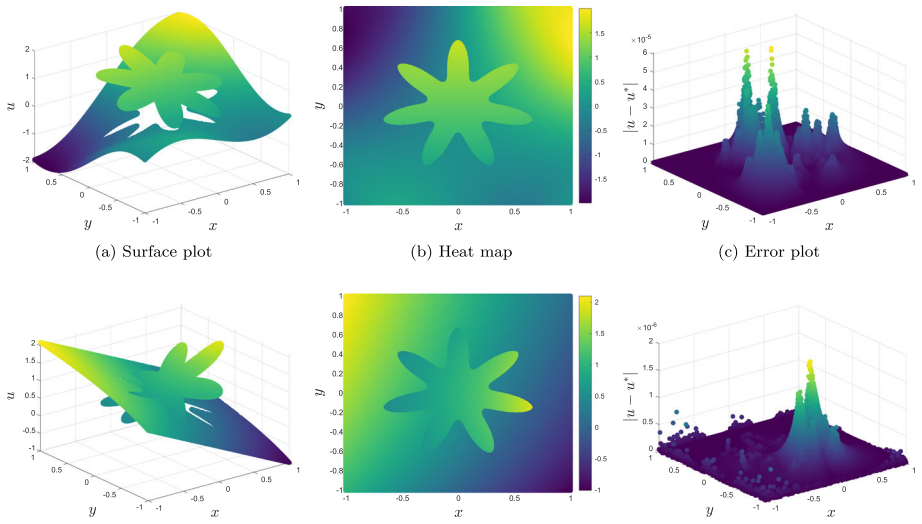
4.4 Example 4: A 2D Diffusion Interface Problem with Multiple Mixed Interfaces

In this example, we consider a diffusion interface problem (1) without reaction in the domain $\Omega = (-1, 1)^2$. The domain Ω is divided by four subdomains Ω_1, Ω_2, Ω_3 and Ω_4 (from the inside to the outside) by the following three closed interfaces:

$$\begin{aligned} \Gamma_1 &= \{(x, y) \mid x^2 + y^2 - 0.2^2 = 0\}, \\ \Gamma_2 &= \{(x, y) \mid x^2 + y^2 - (0.5 - 0.1\sin(5\theta))^2 = 0\}, \\ \Gamma_3 &= \{(x, y) \mid x^2 + y^2 - 0.8^2 = 0\}, \end{aligned}$$

Table 4 Numerical results and running times (measured in seconds) of the proposed RBF-FD method for solving the 2D anisotropic elliptic interface problem in Example 3.

	Example 3(a)						Example 3(b)					
	$(n, p) = (12, 2)$			$(n, p) = (25, 4)$			$(n, p) = (12, 2)$			$(n, p) = (25, 4)$		
N	err_{L2}	Order	Time(s)	err_{L2}	Order	Time(s)	err_{L2}	Order	Time(s)	err_{L2}	Order	Time(s)
15^2	2.97e-2		0.031	2.25e-1		0.058	9.72e-3		0.032	2.31e-3		0.056
30^2	5.79e-3	2.35	0.115	1.26e-2	4.15	0.211	1.68e-3	2.53	0.116	6.18e-5	5.22	0.214
60^2	1.04e-3	2.47	0.451	1.80e-4	6.12	0.844	3.29e-4	2.35	0.467	8.31e-6	2.89	0.867
120^2	3.03e-4	1.77	1.863	9.91e-6	4.18	3.570	8.89e-5	1.88	1.907	3.18e-7	4.70	3.643
Ave. Ord.		**2.19**			**4.81**			**2.25**			**4.27**	



(a) Surface plot (b) Heat map (c) Error plot

(d) Surface plot (e) Heat map (f) Error plot

Fig. 10 Plots of the surface, heat map, and point-wise error maps of numerical solutions by the proposed RBF-FD method with $(n, p) = (25, 4)$ on the densest set of points ($N = 120^2$) in Example 3. Top row: Example 3(a); bottom row: Example 3(b).

Table 5 Numerical results and running times (measured in seconds) of the proposed RBF-FD method for solving the 2D diffusion problem in Example 4.

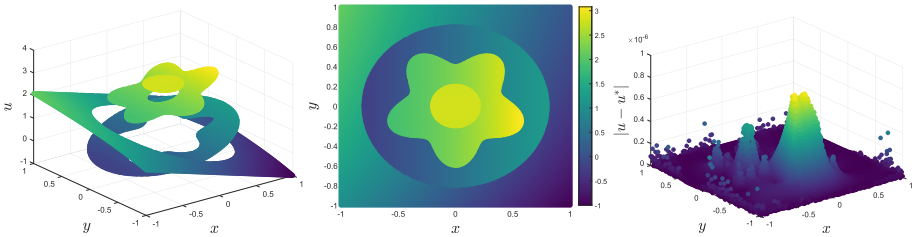
	$(n, p) = (12, 2)$			$(n, p) = (25, 4)$		
N	err_{L2}	Order	Time(s)	err_{L2}	Order	Times(s)
15^2	9.72e-3		0.041	2.31e-3		0.075
30^2	1.68e-2	-0.79	0.135	6.18e-5	5.22	0.249
60^2	1.13e-3	3.89	0.504	8.31e-6	2.89	0.954
120^2	8.89e-5	3.67	2.021	7.42e-7	3.48	3.912
Avg. Ord.		**2.25**			**3.87**	

as illustrated in Figure 6d . The exact solution $u(x, y)$ is given by:

$$u(x, y) = \begin{cases} \cos y + 1.8, & (x, y) \in \Omega_1, \\ \exp(x) + 1.3, & (x, y) \in \Omega_2, \\ \sin x + 0.5, & (x, y) \in \Omega_3, \\ -x + \ln(y + 2), & (x, y) \in \Omega_4. \end{cases} \tag{42}$$

and the diffusion coefficients are chosen as $\Lambda = \sum_{k=1}^{4} \beta_k \xi_{\Omega_k} \boldsymbol{I}$, where ξ_{Ω_k} is the indicator function associated with Ω_k, $\boldsymbol{I}$ is the identity matrix and $(\beta_1, \beta_2, \beta_3, \beta_4) = (1, 10, 100, 0.1)$. The source terms f, boundary conditions g and jump conditions h, φ are determined accordingly.

The corresponding results and running times of the RBF-FD method with same configurations as Example 2 and Example 3 are presented in Table 5 and Figures 11. The numerical



(a) Surface plot (b) Heat map (c) Error plot

Fig. 11 Plots of the surface, heat map, and point-wise error maps of numerical solutions by the proposed RBF-FD method with $(n, p) = (25, 4)$ on the densest set of points ($N = 120^2$) in Example 4.

performance is similar to that of the previous two examples, demonstrating the applicability of the RBF-FD method in handling multiple interfaces problems.

4.5 Example 5: A 3D Anisotropic Diffusion Interface Problem with Two Spherical Interfaces

Finally, we consider a 3D diffusion interface problem (1) without reaction in $\Omega = (-1, 1)^3$ involving two spherical interfaces, defined by

$$\Gamma_1 : \phi_1(x, y, z) = 0, \qquad \Gamma_2 : \phi_2(x, y, z) = 0,$$

where $\phi_1(x, y, z) = x^2 + y^2 + z^2 - 0.3^2$ and $\phi_2(x, y, z) = x^2 + y^2 + z^2 - 0.6^2$. The solution domain Ω is then splited into three parts, namely,

$$\begin{aligned}
\Omega_1 &= \{(x, y, z) \mid \phi_2(x, y, z) < 0\}, \\
\Omega_2 &= \{(x, y, z) \mid \phi_1(x, y, z) > 0,\ \phi_2(x, y, z) < 0\}, \\
\Omega_3 &= \{(x, y, z) \mid \phi_1(x, y, z) < 0\}.
\end{aligned}$$

The exact solution $u(x, y, z)$ and the diffusion tensor $\Lambda(x, y, z)$ are chosen as follow:

$$\Lambda(x, y, z) = \begin{cases}
\begin{pmatrix} 100 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 100 \end{pmatrix}, & (x, y, z) \in \Omega_1, \\
\begin{pmatrix} 1 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{pmatrix}, & (x, y, z) \in \Omega_2, \\
\begin{pmatrix} 100 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 1 \end{pmatrix}, & (x, y, z) \in \Omega_3,
\end{cases} \tag{43}$$

and

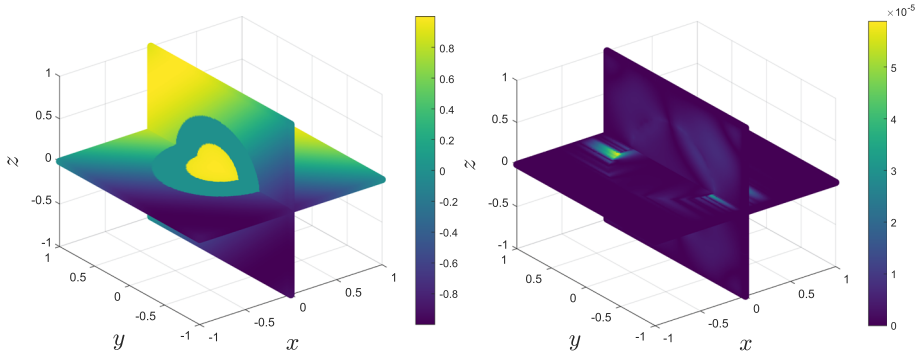
$$u(x, y, z) = \begin{cases}
\sin(x + y + z), & (x, y, z) \in \Omega_1, \\
\sin(x)\sin(y)\sin(z), & (x, y, z) \in \Omega_2 \\
\cos(x)\cos(y)\cos(z), & (x, y, z) \in \Omega_3.
\end{cases} \tag{44}$$

The source term, boundary conditions and jump conditions are then determined accordingly.

We use the RBF-FD method with two configurations: $(n, p) = (25, 2)$ and $(50, 4)$ for most points, except those near the interface or boundary, or on the interface, where $(n, p) = (40, 2)$

Table 6 Numerical results and running times (measured in seconds) of the proposed RBF-FD method for solving the 3D anisotropic diffusion problem in Example 5.

	$(n, p) = (25, 2)$			$(n, p) = (50, 4)$		
N	err_{L2}	Order	Time(s)	err_{L2}	Order	Times(s)
20^3	1.24e-2		2.14	2.56e-3		3.05
40^3	5.43e-3	1.20	9.85	2.80e-4	3.20	12.31
60^3	1.82e-3	2.70	26.63	5.02e-5	4.24	47.50
80^3	1.14e-3	1.63	59.73	1.76e-5	3.64	99.32
Avg. Ord.		**1.84**			**3.70**	



(a) Numerical solution (b) Error map

Fig. 12 Plots of heat map and point-wise error map of the numerical solution (on the planes $z = 0$ and $x = 0$) by the proposed RBF-FD method with $(n, p) = (50, 4)$ on the point set ($N = 60^3$) in Example 4.

and $(80, 4)$ are used, respectively. Note that for 3D problems, $m = 10$ when $p = 2$ and $m = 35$ when $p = 4$, thus the invertibility condition $n \geq m$ for the local linear system is again satisfied for both two configurations. The corresponding numerical results and running times are presented in Table 6, from which we can see that for this 3D example the numerical performance in terms of average convergence rates and running times is similar to that of the 2D case. We also provide additional numerical results, including the heat map and error map of the numerical solution on the planes $z = 0$ and $x = 0$ for the point set with size $N = 60^3$, which are illustrated in Figure 12. Specifically, as shown in Figure 12b , the relative L_2 error for all points remains at the level of $\mathcal{O}(10^{-5})$, demonstrating the accuracy of the proposed method in 3D interface problems.

5 Conclusions

In this paper, we have developed a high-order RBF-FD method for linear anisotropic elliptic interface problems. Through numerical experiments with various interfaces in two and three dimensions, the proposed method has demonstrated high-order accuracy and computational efficiency. It effectively overcomes the challenges associated with traditional mesh generation in handling interface problems. For future research, we plan to extend this work in several directions. Firstly, the necessary conditions for the solvability of the proposed RBF-FD

method still remains an open question (which is a common challenging issue for many RBF-based methods), we will follow up on this topic in the next step. Secondly, we aim to integrate temporal discretization schemes to address time-dependent equations, which will expand the method's applicability to a broader range of problems in science and engineering. Finally, we will explore the combination of the finite difference method and the meshless method, with the goal of significantly reducing computational overhead, thereby enabling the efficient handling of more complex and large-scale problems. By pursuing these advancements, we anticipate that the proposed RBF-FD method will make breakthroughs in theoretical analysis and play a greater role in solving a variety of interface problems in practice.

Data Availability Enquiries about data availability should be directed to the authors.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

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